

# ELI — Gui

ELI v2.1.82

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## ELI GUI

eli/gui/ — 18.3k LOC, 18 files. A full native PySide6/PyQt desktop app (with a Qt-binding compat shim), plus a first-boot launcher and a large scientific “Labs” workspace.

### Tab map (current — 13 main tabs)

flowchart TD  
  W([ELI Pro window]) --> T{13 main tabs}  
  T --> Chat[Chat]  
  T --> Pro[Proactive · 6 sub-tabs]  
  T --> Img[Images]  
  T --> QA[Quick Actions]  
  T --> Scr[Screen]  
  T --> Fil[Files]  
  T --> Cod[Coding]  
  T --> Tsk[Tasks · all background jobs]  
  T --> Rep[Report Builder]  
  T --> Exp[Experimental · read-only workbench]  
  T --> Wld[Eli's World]  
  T --> Set[Settings · 10 pages: Model · Runtime · Generation · Identity · Audio · Application ·  
  T --> Labs[Labs · 10 sub-tabs]  
    Labs --> L1[Notebook · Memory · Jupyter · Calculator · Physics]  
    Labs --> L2[File Chat · Workspaces · Sim/IDE]  
    Labs --> L3[Orchestration · Test & Review<br/>dev/diagnostic tools]

## Files

| File                             | LOC   | Role                                           |
|----------------------------------|-------|------------------------------------------------|
| eli_pro_audio_gui_v2_0.py        | 12.1k | the main window + most app logic (god-file)    |
| labs_tab.py                      | 5.7k  | scientific workspace tab                       |
| app.py                           | 817   | launcher / first-boot auto-tune / entry main() |
| panels/startup.py                | 1305  |                                                |
| panels/settings.py               | 723   |                                                |
| docks/operator_console_dock.py   | 303   | operator console dock                          |
| widgets/ollama_model_selector.py | 294   | optional Ollama model picker                   |
| tabs/experimental_tab.py,        | small | tabs/docks/widgets + Qt compat                 |
| panels/agent_wizard.py,          |       |                                                |
| docks/proactive_dock.py,         |       |                                                |
| tabs/eli_world_tab.py,           |       |                                                |
| qt_compat.py, panels/_qt.py      |       |                                                |

## Launcher (app.py)

The entry path: `_detect_hardware` (queries **free** VRAM — the display server consumes VRAM before ELI launches, so `free`  $\neq$  `total`), `_auto_tune(model_path, hw)` (picks `n_ctx`/layers/batch), `_pick_model`, `_confirm_params`, config load/save. `main()` either shows the startup model picker (first boot) or delegates to `eli_pro_audio_gui_v2_0.main()`. `--setup` forces the wizard.

## Main window (eli\_pro\_audio\_gui\_v2\_0.py)

A 11.0k-line module holding the window **and** a stack of embedded classes that are really application logic, not just UI: - `CentralMemoryAdapter` — bridges the GUI to the memory subsystem. - `LocalModelManager` (708) — discover/load/swap local GGUF models. - `OllamaModelManager` (1142) — optional Ollama integration (legacy/optional; ELI's stance is 100% local GGUF, so this is a secondary path). - `ExecutorBridge` (1246) — routes GUI actions into the executor/engine. - `_GUIEngineAdapter` — engine façade for the UI. - UI widgets: `_QABoard` (quick-action card board), `_MiniTelemetryGraph` (live telemetry), `_ZoomableSettingsView`, `_ZoomableImagePreview`, `_FlowLayout`, `_CapabilityList`. - `pyqtSignal/Slot` aliased through `qt_compat.py` so it runs on PyQt **or** PySide.

## Labs workspace (labs\_tab.py)

A 5.7k-line “scientific workspace” tab with **10 sub-tabs**: Notebook, Memory & Conversations, Jupyter, Calculator, Physics constants, File Chat, Workspaces, Sim/IDE, **Orchestration**, **Test & Review**. (Report Builder was promoted out of Labs to its own main tab; Orchestration + Test & Review were demoted back INTO Labs in the 2026-06-18 advisory as developer/diagnostic tools.) This is the research-bench surface (reflects whatever technical/research work the active user does, surfaced dynamically from their own data).

## Other surfaces

- `panels/startup.py` — guided first-boot: detect hardware → pick/download a GGUF (HuggingFace) → tune params.
- `panels/agent_wizard.py` — author custom agents (writes to the trust-gated custom-agents dir; see `security.md`).
- `docks/operator_console_dock.py`, `docks/proactive_dock.py` — operator console
  - proactive-suggestion dock.
- `widgets/ollama_model_selector.py`, `tabs/experimental_tab.py`, `tabs/eli_world_tab.py` — optional/experimental surfaces.

## Honest assessment

- **Strong:** this is a genuine, feature-rich desktop product — dockable panels, quick-action board, live telemetry graph, zoomable settings/image preview, local model management, a first-boot wizard, an agent-authoring wizard, and a full scientific workspace. Cross-binding (PyQt/PySide) compat is handled. Most local-LLM projects ship a chat box; this is an application.
- **Weak / watch:**
  1. **God-file #3** — `eli_pro_audio_gui_v2_0.py` (11.0k) mixes UI with core logic (`LocalModelManager`, `ExecutorBridge`, `CentralMemoryAdapter`, `_GUIEngineAdapter`). The model/executor/memory bridges should live outside the window module so the UI isn't coupled to core internals (and so they're testable headless). `labs_tab.py` (5.7k) is a second large file.
  2. **Ollama manager** (1.1k LOC) sits oddly against the "100% local GGUF, don't-care-about-Ollama" stance — it's an optional/legacy path carrying real weight; candidate for removal or clear quarantine.
  3. UI logic instantiating engine/memory directly makes a clean headless mode harder (there is `eli --headless`, but the GUI module re-implements bridges rather than sharing one service layer).